IRexp experimental infrared band lists from the PMC Open Access Subset and Chemotion

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Abstract

IRexp is a redistributable collection of **experimental infrared band lists** (cm^{-1} peak positions) mined from open chemistry literature, optionally with $^{1}H/^{13}C$ shifts and resolved structures. The release holds **121,233** records (119,345 PMC OA; 1,888 Chemotion/RADAR4Chem), with **43,060** structure-linked and **33,201** full IR + ^{1}H + ^{13}C + structure quadruples. IRexp stores **numeric band lists**, not absorbance traces. Records carry `source_doi` and stamped `license` / `license_pool`. Licensing is **mixed**: Europe PMC join yields **87,617** commercial (CC-BY/CC0), **20,938** non-commercial (NC*), **10,781** empty/unknown (excluded from commercial Zenodo), and **1,897** ShareAlike (Chemotion + rare PMC SA). Reuse: multimodal training and complementary elucidation benchmarks citing this Descriptor. Data: Hugging Face `ilkhamfy/IRexp`; Zenodo DOI pending archival mint. Code is MIT-licensed.

1 Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet **open, redistributable** collections of *experimental* IR data remain sparse relative to modern machine-learning needs. Digitised absorbance libraries such as the NIST Chemistry WebBook [National Institute of Standards and Technology, 2026] and AIST SDBS [National Institute of Advanced Industrial Science and Technology, 2026] are valuable but either modest in size or **view-only** (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra [Zipoli et al., 2025], while large literature mines for **NMR** peak lists (notably NMRexp [Wang et al., 2025]) demonstrate that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans [Yang et al., 2026]) further motivate an **IR-focused, redistributable** band-list resource rather than another closed spectrum archive.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound **IR band lists** (wavenumbers in cm^{-1}) together with $^{1}H/^{13}C$ NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a **different object** from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the **PMC Open Access Subset** bulk S3 mirror and peak lists from the **Chemotion** FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).
3. Resolves compound names to canonical structures with OPSIN [Lowe et al., 2011], RDKit [Landrum, 2026], and SELFIES [Krenn et al., 2020] where possible.
4. Releases **extracted numbers only** — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IRexp is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IRexp holds structure-linked band lists. The intended reuse is multimodal pretraining, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — **not** reproduced here) [Yabbarov and Vargas-Hernández, 2026].

What this Data Descriptor does not contain. No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

2 Methods

2.1 Source corpora

PMC Open Access Subset. Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`) [National Library of Medicine, 2026]. **Harvest window (recoverable from git snapshots):** the bulk S3 IR crawl and `seen_papers` / `ir_harvest_snapshot` artefacts were produced on **2026-06-07 (UTC)** (`scripts/s3_ir_harvest.py`; incremental auto-snapshots that day through ∼134,893 raw IR rows). Chemotion was ingested the same day. A harvest snapshot records **188,016** distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from **15,416** unique `PMC:*` accessions. Extraction operates on open-access **plain text** objects at flat S3 keys `PMC{id}.{v}/PMC{id}.{v}.txt`. Europe PMC full-text XML is used for some validation re-fetches. The **released** IRexp DOIs are exclusively `PMC:*` accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

Discovery vs `oa_comm` / `oa_noncomm` packages. Identifier discovery used NCBI E-utilities `esearch` over PMC with an IR-characterisation query **and** `open access[filter]`, then fetched matching IDs from the flat S3 text layout. The harvest therefore did **not** pre-filter by walking the PMC OA Subset's `oa_comm` vs `oa_noncomm` vs other package directories. PMC OA is still **not** a single licence [National Library of Medicine, 2026]. Licence truth for redistribution is applied **post-hoc**: every unique PMCID in IRexp (15,416) was joined to Europe PMC `license` metadata (`scripts/join_pmc_licences.py`); each record carries `license` / `license_pool`. Commercial-use (CC-BY/CC0) rows form the Zenodo / Sci Data primary pool; NC* are held aside; empty/unknown are **excluded** from the commercial deposit — aligning redistributable intent with the OA commercial/non-commercial distinction via article-level licences rather than S3 package paths (`data/NOTICE`; `docs/scientific_data/LICENCE_REMEDIATION.md`).

Chemotion / RADAR4Chem. **1,888** records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI `10.22000/OGoEQGlsZGElrgst`) [Jung et al., 2024], licensed **CC-BY-SA-4.0**. Ingest (`scripts/chemotion_to_irexp.py`, 2026-06-07): download the MD5-verified deposit; for each of **2,116** ATR-IR analyses, flatten the Quill-delta `content` field to plain text; parse an **author-curated** IR band list with the **same** regex extractor and quality gates as the PMC path; resolve the deposit's canonical SMILES with RDKit → InChIKey + SELFIES; keep the richest band list per InChIKey; drop the **8** molecules already present in the PMC pool → **+1,888** new structure-resolved rows. These are **not** algorithmic peak-picks from absorbance curves; they are author-entered experimental band lists from the

ELN deposit, denser than typical paper prose (median **39** bands vs **9** for PMC). Rows carry `license=CC-BY-SA-4.0` and `source_doi` equal to the deposit DOI.

Excluded. AIST SDBS (view-only; no bulk export) [National Institute of Advanced Industrial Science and Technology, 2026]; NIST WebBook join code exists in the repository but contributes **0** records to the released IRexp (`ir_source` is always `experimental` for released rows).

2.2 Discovery and fetch (released corpus)

1. Discover IR-reporting OA PMCIDs via NCBI `esearch` (`open access[filter]` + characterisation-format IR query; year/month sliced).
2. Fetch OA full text from PMC-OA S3 plain-text objects (`PMC{id}.{v}.txt`).
3. Ingest Chemotion deposit author-curated band lists + structures (`scripts/chemotion_to_irexp.py`).
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl.gz` (134,893 rows including intermediate prose fields; the curated release is `irexp.jsonl.gz`).

Non-OA / Scrapling fence. Development adapters for ChemRxiv, Beilstein, and generic publisher pages exist for exploration. They are **outside the construction path of the released dataset**. **No released `source_doi` is a paywalled publisher DOI.** Methods for IRexp as published here cite only **PMC-OA S3 + Chemotion**.

2.3 Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers → `ir_bands_cm-1` (list of floats, cm^{-1});
- ^1H and ^{13}C payloads → `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives. **No curve digitisation** is performed: figures are not traced; JCAMP-DX is not stored.

2.4 Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN [Lowe et al., 2011], canonicalised with RDKit [Landrum, 2026], and encoded as InChIKey and SELFIES [Krenn et al., 2020]. An optional PubChem [Kim et al., 2023] name fallback (`USE_PUBCHEM`) handles trivial names. Structure coverage of the full release is **35.5%** (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

2.5 Licence handling

Pool	Records	Stamp
commercial (CC-BY + CC0)	**87,617**	`license_pool=commercial`; Zenodo / Sci Data primary
non_commercial (CC-BY-NC*)	20,938	held aside
sharealike (Chemotion + rare PMC SA)	1,897	CC-BY-SA / CC-BY-SA-4.0
empty_unknown	10,781	excluded from commercial deposit

`scripts/join_pmc_licences.py` stamps every row; `scripts/split_license_pools.py` reports provenance and materialises pool files under `data/irexp/licence_pools/`. Narrative and policy: `docs/scientific_data/LICENCE_REMEDIATION.md`.

2.6 Quality tooling

Optional physics gates live in `spectro_scraper/quality.py` (^{13}C peak count $\leq$ carbon count; ^{1}H integration vs formula; IR wavenumber windows). They were **not** applied as a hard filter at harvest; Technical Validation reports a full-corpus post-hoc quarantine on `irexp_resolved` (`scripts/quarantine_structure_nmr.py` $\rightarrow$ `data/audit/structure_nmr_quarantine.jsonl.gz`). Transcription and recall-proxy scripts: `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`.

3 Data Records

3.1 Object definition

Each IRexp record is a JSON object. Required chemistry fields for an IR entry:

Field	Type	Description
`id`	string	Stable record id (InChIKey when resolved, else internal)
`ir_bands_cm-1`	float list	Experimental IR peak positions (cm^{-1})
`ir_source`	string	`experimental` for all released rows
`source_doi`	string	`PMC:<pmcid>` or Chemotion deposit DOI
`h_nmr` / `c_nmr`	string or null	Author-reported shift lists
`smiles` / `selfies` / `inchikey`	string or null	Resolved structure encodings
`has_structure`	bool	Convenience flag
`license` / `license_pool`	string	Per-article stamp (Europe PMC join or Chemotion)

Not included: absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

3.2 Files and counts

Paths relative to the project repository / Hugging Face mirror.

File	Records	Description
`data/irexp/irexp.jsonl.gz`	121,233	Full curated release
`data/irexp_resolved/irexp_resolved.jsonl.gz`	43,060	100% structure-linked
`data/irexp_release/train_no_bench.jsonl.gz`	42,808	Resolved minus IRSpectra-Bench InChIKeys
`data/irexp_release/train_no_bench_nmr.jsonl.gz`	32,949	Same with both ^{1}H and ^{13}C
`data/irexp_release/pretrain_ir.jsonl.gz`	119,345	PMC-only IR pretrain pool
`data/irexp/seen_papers.txt.gz`	188,016 lines	PMC IDs scanned at harvest
`data/irexp/ir_harvest_snapshot.jsonl.gz`	134,893	Intermediate harvest snapshot
`licence_pools/irexp_commercial.jsonl.gz`	87,617	CC-BY + CC0 (Zenodo/Sci Data primary)
`licence_pools/irexp_non_commercial.jsonl.gz`	20,938	NC* held aside
`licence_pools/irexp_sharealike.jsonl.gz`	1,897	Chemotion + rare PMC SA
`licence_pools/irexp_empty_unknown.jsonl.gz`	10,781	Excluded from commercial

Composition of `irexp.jsonl.gz`.

Slice	Count
All IR band-list records	121,233
With ^{1}H and/or ^{13}C NMR	87,075 (72%)
With resolved structure	43,060 (35.5%)
Structure + any NMR	40,702
Full IR + ^{1}H + ^{13}C + structure	33,201
PMC OA provenance	119,345
Chemotion provenance	1,888
Unique PMC accessions	15,416

Licence-pool counts (confirmed Track 1 join, 2026-08-26). Lookup over **15,416** unique PMCIDs; stamped pool files under `data/irexp/licence_pools/`. Sum of pools = **121,233** (unchanged).

Pool file	Count	Notes
`irexp_commercial.jsonl.gz`	**87,617**	CC-BY / CC0 — **Zenodo primary**
`irexp_non_commercial.jsonl.gz`	**20,938**	CC-BY-NC* held aside
`irexp_empty_unknown.jsonl.gz`	**10,781**	empty / unresolved — **excluded** from commercial deposit
`irexp_other.jsonl.gz`	**0**	reserved (e.g. CC-BY-ND)
`irexp_sharealike.jsonl.gz`	**1,897**	Chemotion CC-BY-SA-4.0 (1,888) + rare PMC SA

The full `irexp.jsonl.gz` remains multi-licence on disk; commercial redistribution must use the commercial pool (or `license_pool == "commercial"`). Hugging Face was re-issued with stamped pools and honest card text (`scripts/publish_hf.py`, 2026-08-26).

Median bands: **9** (PMC), **39** (Chemotion). All **1,360,866** released IR band values fall inside 350–4000 cm^{-1} (full-corpus range check; Technical Validation).

3.3 Access

- **Hugging Face:** https://huggingface.co/datasets/ilkhamfy/IRexp
- **GitHub development mirror:** https://github.com/IlkhamFY/spectro-agent
- **Zenodo archival snapshot:** DOI pending mint after honest multi-licence metadata.

4 Technical Validation

Machine-readable package: `docs/scientific_data/qc_structure_nmr.json` (and artefacts under `data/audit/`).

4.1 Transcription fidelity (PMC)

On a seed-fixed random sample of **60** PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text ($\pm 1\,\mathrm{cm}^{-1}$ integer match). Result: **560/560 bands** and **60/60 records** confirmed. Enlarged sample **n=200** (same script/seed family, `data/audit/extraction_audit_n200.json`): **2,250/2,261 bands (99.51%)** and **196/200 records (98.0%)** fully confirmed. This bounds **transcription** error (hallucinated, mis-parsed, or unit-mangled values).

4.2 Extraction-recall proxy (PMC, harvest path)

A human mark-up of every IR string in every paper remains the gold standard. As an automatic **proxy** on the actual harvest path (`scripts/audit_extraction_recall.py --n 40 --seed 0`): re-fetch PMC-OA S3 plain text for **40** distinct source PMCIDs; re-run `extract_records`; compare band-sets to released rows ($\pm 1\,\mathrm{cm}^{-1}$). Result: **2,640/2,640** released bands confirmed in re-extract; **248/248** released IR lists recovered (list-level recall proxy **1.0**; 40/40 papers); **4/40** papers yielded *extra* re-extracted lists relative to the curated release (parser over-fire vs post-harvest dedup/curation, not missed released compounds). Summary: `data/audit/extraction_recall_proxy.json`. This is **not** a substitute for expert human recall.

4.3 Structure–NMR consistency and quarantine (resolved)

Sample (prior). On **500** `irexp_resolved` records with ^{13}C text (seed 0): **17/500 (3.4%)** listed more peaks than carbons. On **500** with ^{1}H text: integrals > formula H+2 in **17/497 (3.4%)**.

Full resolved corpus. `scripts/quarantine_structure_nmr.py` applied the same physics gates to all **43,060** structure-linked rows. **1,882 (4.37%)** fail $\geq$1 hard check and are listed in `data/audit/structure_nmr_quarantine.jsonl.gz` (diagnostic only — release files unchanged). Among rows with the relevant modality: ^{13}C peaks > carbons **1,194/34,231 (3.49%)**; ^{1}H integral > formula+2 **1,141/39,672 (2.88%)**; IR out-of-range **0**; unparseable SMILES **0**. Re-users should filter quarantined IDs before supervised training. These rates are integrity diagnostics, **not** an expert skeleton audit.

4.4 IR physical window

Every band in the full 121,233-record release lies in **[350, 4000] $\mathbf{cm^{-1}}$** (0 / 1,360,866 out of range).

4.5 QC status

Check	Status
Per-PMCID licence join + pool counts	**Done** — commercial 87,617
Transcription fidelity n=60 and n=200	**Done**
Extraction-recall automatic proxy (n=40 papers, S3 path)	**Done** (human recall still optional)
Full-corpus structure–NMR quarantine	**Done** — 1,882 / 43,060 flagged
Expert structure spot-check ($n \geq 100$)	Optional / deferred

5 Usage Notes

- **Band lists $\neq$ spectra.** Do not evaluate models trained on IRexp as if they had seen full absorbance curves.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, author-curated ELN band lists). Keep Chemotion ShareAlike constraints in mind when combining pools; combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0.
- **Do not assume PMC = CC-BY-4.0.** Filter to `license_pool == "commercial"` (or use `irexp_commercial.jsonl.gz`) for commercial redistribution; attribute via `source_doi` / `pmcid`.

- **Structure–NMR quarantine.** Before supervised training on `irexp_resolved`, optionally drop IDs in `data/audit/structure_nmr_quarantine.jsonl.gz` (∼4.4% of resolved rows).
- **Training without benchmark leakage.** If using the complementary IRSpectra-Bench problems, fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld.
- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI and attribute originating articles through each record's `source_doi`.

6 Data Availability

IRexp numeric extracts are available at:

- Hugging Face Datasets: https://huggingface.co/datasets/ilkhamfy/IRexp
- GitHub: https://github.com/IlkhamFY/spectro-agent (`data/irexp/`, `data/irexp_resolved/`, `data/irexp_release/`)
- Zenodo: archival DOI at proof; primary artifact = commercial pool, `cc-by-4.0` metadata + SA companion

Licensing summary (honest):

- **Chemotion (1,888):** CC-BY-SA-4.0 [Jung et al., 2024].
- **PMC (119,345):** mixed Creative Commons — stamped per article; commercial redistributable **87,617** (CC-BY/CC0); NC* **20,938** held aside; empty/unknown **10,781** excluded from commercial Zenodo (`LICENCE_REMEDIATION.md`).
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

7 Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in https://github.com/IlkhamFY/spectro-agent under the **MIT License** (`LICENSE`). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); release harvest `scripts/s3_ir_harvest.py`, `scripts/chemotion_to_irexp.py`; validation `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`, `scripts/quarantine_structure_nmr.py`; pool split `scripts/split_license_pools.py`. The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

8 Author contributions

I.Y.: conceptualization, methodology, software, data curation, validation, writing — original draft.
R.A.V.-H.: conceptualization, supervision, writing — review and editing.

9 Competing interests

The authors declare no competing interests.

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